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Explainability Of Artificial Intelligence For Diagnosing COVID-19 From Chest X-Rays

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Abstract—**This COVID-19 pandemic has overburdened the government and the healthcare system of many countries around the world. It has brought up the need for a fast and accurate diagnosing method. Artificial intelligence (AI) is having a notable role in different aspects of the pandemic-contact tracing, epidemiology, medical diagnosis and prognosis, and drug development. Deep learning has found its application in the diagnosis of COVID-19 chest X-rays (CXR) using convolution neural nets. Many architectures have been used and transfer learning is the most preferred approach. These models have proven to be fast and accurate in COVID-19 diagnosis. However, one key element that has prevented the use of AI in clinical practice is its lack of transparency and explainability. In this paper, we use the ResNet-50 pre-trained model for classifying the CXR of COVID-19 patients from pneumonia and normal patients. We then use explainability algorithms to visualize the model features and verify the explainability of the model.**

Keywords—COVID-19, coronavirus, X-ray, transfer learning, diagnosis, artificial intelligence, explainability

I. INTRODUCTION

The fight against the COVID-19 pandemic has brought the focus back to public health policies. Until a worldwide mass vaccination is carried out, increased testing and contact tracing is the key strategy to prevent the coronavirus from spreading and identifying hotspots.

A. Need for More Accurate Diagnosing

The gold standard for diagnosing COVID-19 is the reverse transcriptase-polymerase chain reaction (RT-PCR) test. It uses respiratory specimens to detect SARS-CoV-2 RNA. Although RT-PCR is highly accurate and trusted, it is a complicated and time-consuming task. Also, the kits are limited in supply. Studies have shown the RT-PCR test to have a highly variable sensitivity [1]. Moreover, depending on how the specimen was collected, RT-PCR has shown to have a fluctuating positive rate [2]. Other molecular tests include antigen tests but they tend to have a high false-negative rate. Serological tests detect antibodies in the body but it takes many days, even weeks, to develop antibodies after an infection and hence aren't much reliable [3].

B. Potential Role of Artificial Intelligence

The Radiology examination using chest radiography is another key screening approach for COVID-19. These include computed tomography (CT) and X-ray scans. However, because the visual cues can be quite difficult to identify, one major bottleneck faced is the availability of expert radiologists to read these radiological scans.

Artificial intelligence (AI) can play a significant role in such scenarios. This project focuses on using deep neural networks for automating the diagnoses of COVID-19

II. LITERATURE REVIEW

Many AI models have been proposed for the diagnosis of COVID-19 and other lung diseases. Most of them use the concept of transfer learning. Pretrained networks like Densenet [4], Resnet [5], and VGG [6] are modified and trained using labeled radiological images. COVID-Net is a network designed specifically for COVID-19 classification [7]. Table I shows a comparison of various models proposed.

A. Preference for X-rays

Most of these studies used CT volumes. But CT scanners are costly and not available everywhere. The decontamination time between the scans is long. Moreover, they expose patients to high doses of radiation. Hence, the use of CT scans for COVID-19 diagnosis has been limited. X-rays are routine, easily available, and cost-effective. Also, the easy portability of X-ray machines makes it a much better option for large-scale testing.

B. Explainability of Artificial Intelligence

AI is seen as a black box, as it's hard to understand its reasoning. This opacity has limited its application in healthcare. It's hard to rely on AI models for critical decisions. A sample of a dataset used in many studies for training AI models is shown in fig. 1 [8]. Some studies reported accuracy of more than 90%. However, the dataset actually used pediatric chest X-rays for pneumonia samples and adult chest X-rays for normal samples. Hence, there is a possibility that the model may have learned the wrong features for classification. This highlights the importance of explainable AI.

Explainability algorithms can be used to validate if the decisions made by the model are based on correct information instead of inappropriate information (e.g. embedded markup symbols, false visual indicators apart from the area of interest, imaging artifacts, etc). Without an efficient explainability-driven auditing strategy, such scenarios of correct decisions but faulty reasonings are very difficult to identify and track. This is important, especially in life-death scenarios.

Gradient-weighted Class Activation Mapping (Grad-CAM) [9], Occlusion Sensitivity [10] and Locally Interpretable Model-agnostic Explanations (LIME) [11] are some

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TABLE I. COMPARISON OF AI MODELS FOR DIAGNOSIS OF COVID-19

Modality	Subjects	Architecture	Accuracy (%)	Parameters (10^6)	Study
X-ray	389 Covid-19 + 1642 normal + 1642 Pneumonia	ResNet-50	89.78	24.7	Our study
X-ray	125 Covid-19 + 500 Pneumonia + 500 No-Findings	DarkCovidNet	87.02	1.1	Ozturk et al. [12]
CT	219 Covid-19 + 500 Influenza-A viral Pneumonia + 175 Healthy	ResNet-18	86.7	11	Butt et al. [13]
X-ray	224 Covid-19 + 700 Pneumonia + 504 Healthy	VGG19	93.48	143	Apostolopoulos and Mpesiana [14]
X-ray	284 Covid-19 + 310 normal + 657 Pneumonia	FC-DenseNet/ResNet-18	88.9	11.6	Oh et al. [15]
X-ray	284 Covid-19 + 310 normal + 657 Pneumonia	CoroNet	94.59	33	Khan et al. [16]
CT	510 Covid-19 + 510 Non-Covid-19	ResNet-101	99.51	44.6	Ardakani et al. [17]
CT	510 Covid-19 + 510 Non-Covid-19	Xception	99.02	22.9	Ardakani et al. [17]
X-ray	266 Covid-19 + 8066 normal + 5538 Pneumonia	COVID-NET	93.3	11.75	Wang et al. [7]

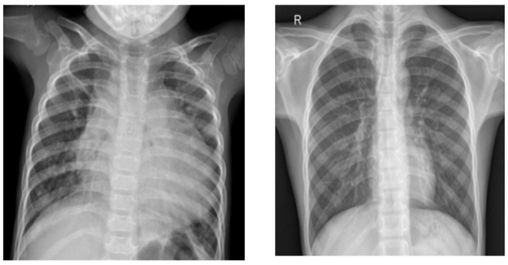


Fig. 1. Example of erroneous dataset leading to biased models. The left image is a pneumonia sample and the right image is a normal sample.

of the explainability algorithms that can be used to visualize the features and explain the prediction of any classifier. Apart from building trust, identification of the vital features leveraged by the AI model and their visualization can aid physicians discover new important visual markers associated with the COVID-19 infection. This information can help further improve the accuracy of diagnosis.

III. OBJECTIVE

The motivation for this study was to build a robust AI model that can help physicians better decide who should be prioritized for confirmation of COVID-19 infection by an RT-PCR test. Also, pneumonia and COVID-19 differentiation is important to prevent a twindemic of flu and coronavirus.

The study focuses on the explainability of AI models used for medical diagnoses. Furthermore, insights from the explanations can help radiologists find new visual indicators for the diagnosis of COVID-19.

IV. METHODOLOGY

A. Dataset Formulation

Various online datasets were explored and three trusted sources were selected to formulate a single labeled dataset :

1) COVID-19 image data collection on GitHub [18].

2) RSNA pneumonia detection challenge [19].

3) NIH chest X-rays [20].

The dataset was modified to represent equal male and female chest X-rays (CXR) in each class. The final dataset comprised of 389 COVID-19 CXR, 1642 pneumonia CXR, and 1642 normal CXR. The data was split in the ratio 8:2 into train-set and test-set. Fig. 2 shows a few samples from the dataset.

B. Transfer Learning

Various pre-trained networks are open-sourced. These deep models have rich feature representation as they have been trained on very large datasets. Using transfer learning to fine-tune these networks is a simple and faster approach than training a completely new model with randomly initialized weights. The learned parameters can be efficiently transferred to a new task requiring a much lesser number of training images. We used ResNet-50 for this study that has been pre-trained on the ImageNet dataset comprising of 1.2 million images. The architecture consists of 48 Convolution layers, 1 MaxPool, and 1 Average Pool layer.

C. Explainability Algorithms

We used 3 algorithms to visualize the model features and explain the model predictions :

1) *LIME:* The algorithm segments an image into its features. Then synthetic images are generated by excluding a few features randomly. The excluded features are replaced with the mean pixel value. This synthetic dataset is classified using the deep network. Then a simpler regression model, e.g. regression tree, is fitted using the presence/absence of image features as the binary feature matrix and the classification scores for the target class as the target vector. The importance of the features is hence computed using the regression model, which is then converted into a map indicating the most important features in an image.

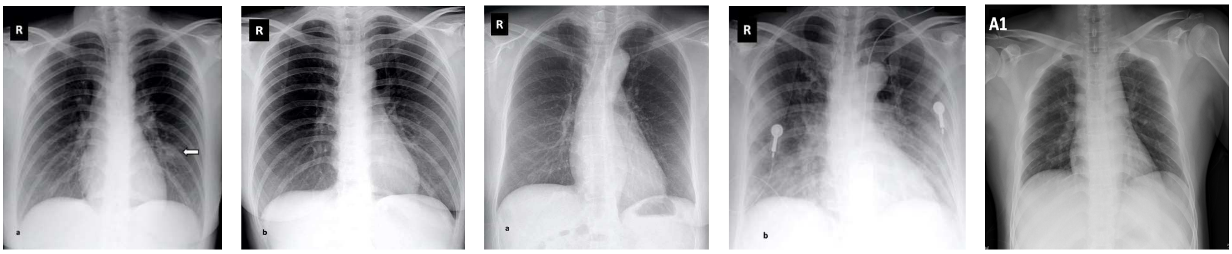


Fig. 2. Samples from the dataset

2) *GradCAM:* It is a generalization of class activation mapping (CAM). To identify the most important features of the image for classification, it calculates the derivative of the classification score with respect to the final convolutional feature map of the network.

3) *Occlusion Sensitivity:* The algorithm substitutes a block of the image with a grey occluding mask which moves across the image with each iteration hence causing distortion in different regions of the image. The change in classification score for the target class is calculated with respect to the coordinates of the mask. When an important feature is occluded in the image by the mask, the score drops considerably.

V. Results

Using Resnet-50 for transfer learning, a model with 89.78% accuracy was built. The training process is shown in fig. 3. Fig. 4 shows the confusion matrix for the test set. Fig. 5 shows the heat maps from the 3 explainability algorithms along with the original samples. Randomly, 2 COVID-19 samples, 2 pneumonia samples, and 2 normal samples from the test set are shown for comparison. The features highlighted in red have increased the probability for the target class the most. The features highlighted in blue have least affected the classification. For most of the samples, the red regions are within the area of interest, i.e. lungs. However, for a few samples, like COVID-19 sample-2, many features are outside the area of interest. Hence, the explainability of the AI model couldn't be established satisfactorily.

VI. Discussion And Conclusion

The guidelines from the Centers for Disease Control and Prevention (CDC) state that CXR and CT should not be used to diagnose COVID-19 [21]. However, some studies have suggested that radiological imaging like CT may be just as sensitive as RT-PCR [22]. Various studies have reported deep neural net outperforming expert radiologists in many diagnosing tasks [23]. Hence this study tries to explore the scope of AI in assisting with the fight against COVID-19.

From the results, we conclude that AI models can classify radiography images with an accuracy comparable to radiologists. However, the explainability algorithms couldn't explain all the model predictions satisfactorily. Hence, further research and improvement is required. These

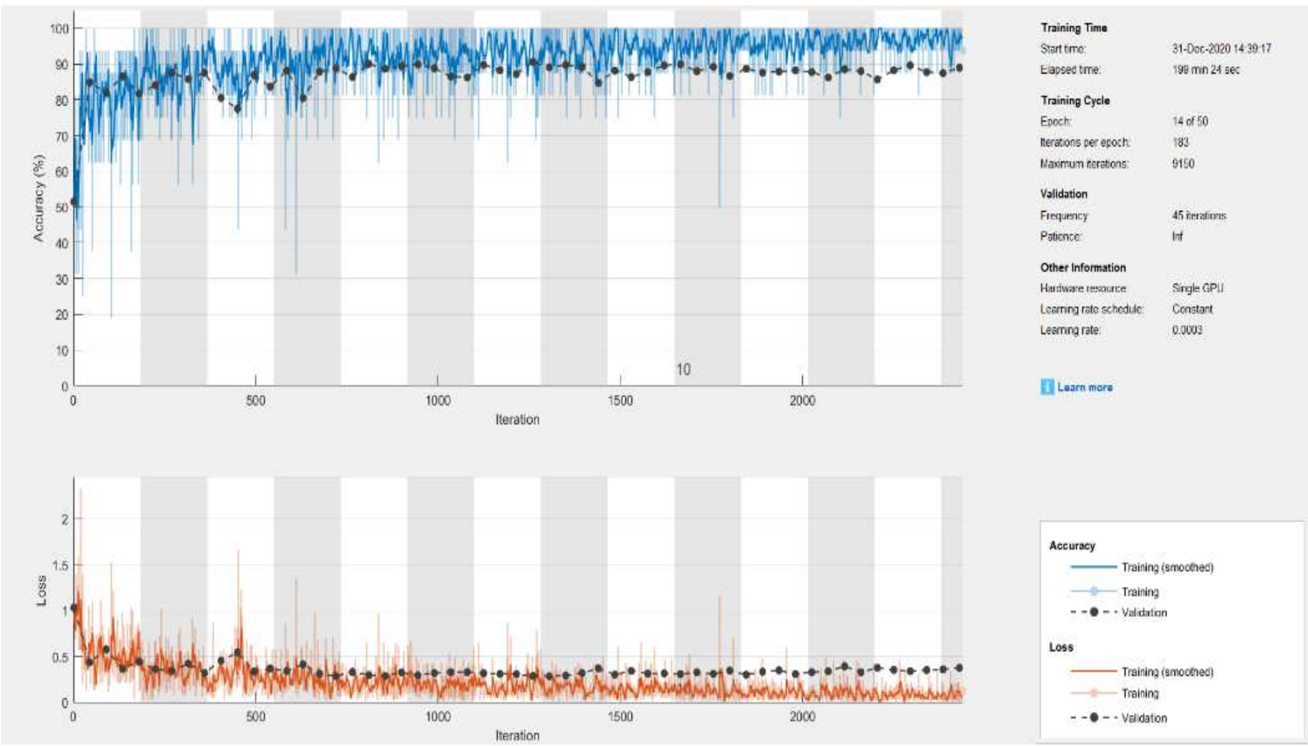


Fig. 3. Training process for ResNet-50 model.

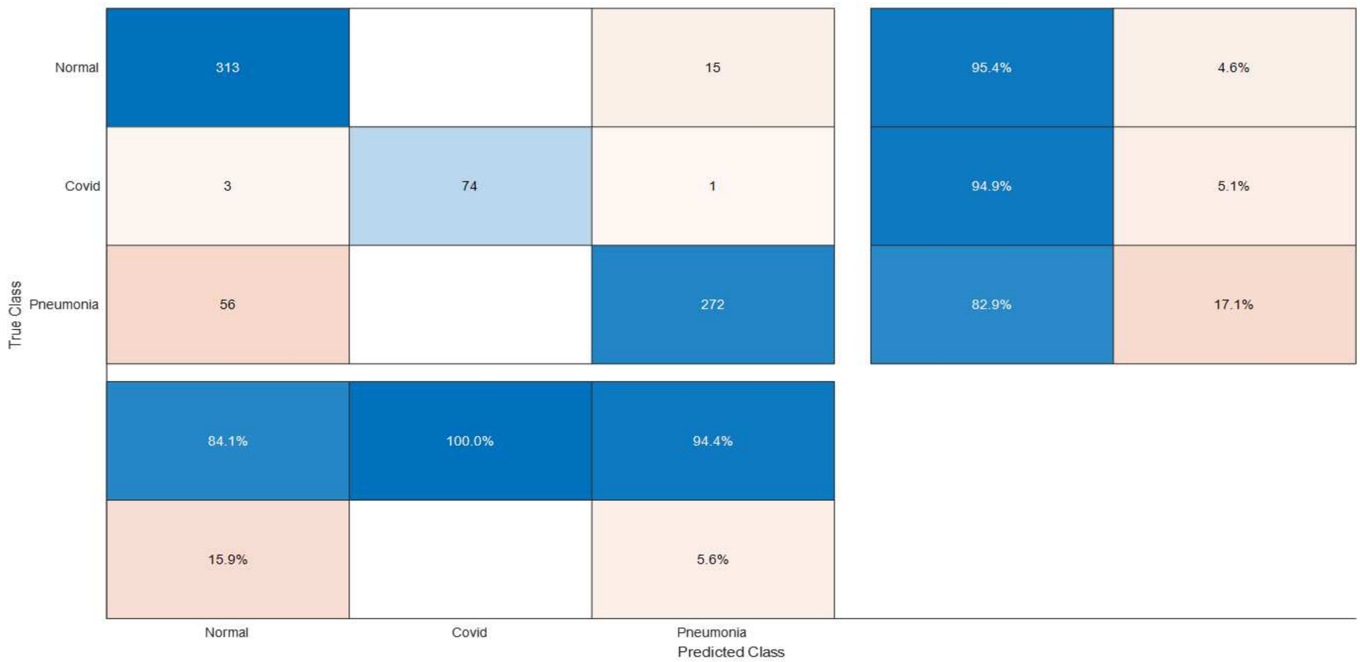
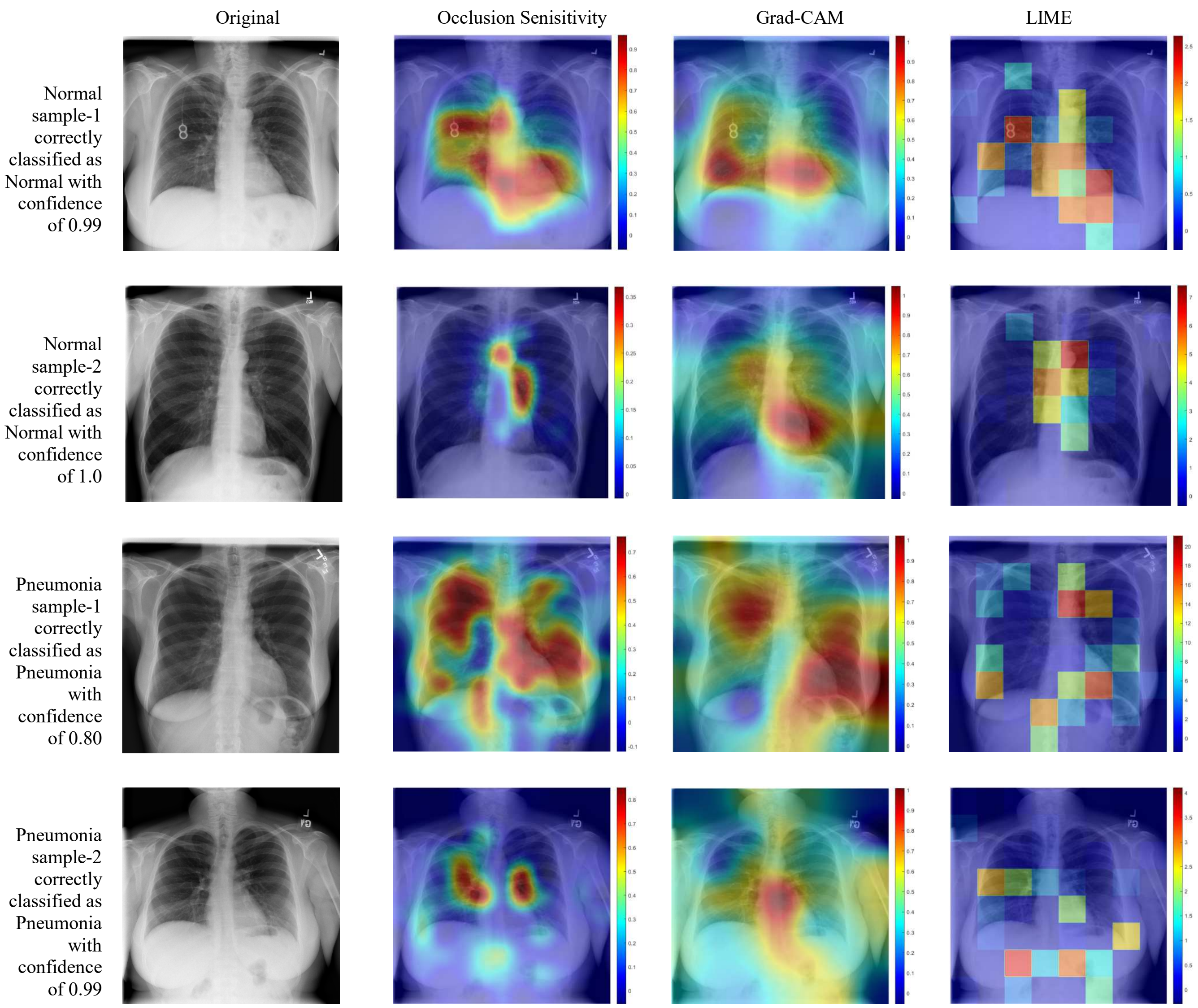


Fig. 4. Confusion matrix, sorted according to decreasing class-wise recall (row summary, right). Column summary (bottom) shows the class-wise precision.



COVID-19 sample-1 correctly classified as COVID with confidence of 0.86

COVID-19 sample-2 correctly classified as COVID with confidence of 1.0

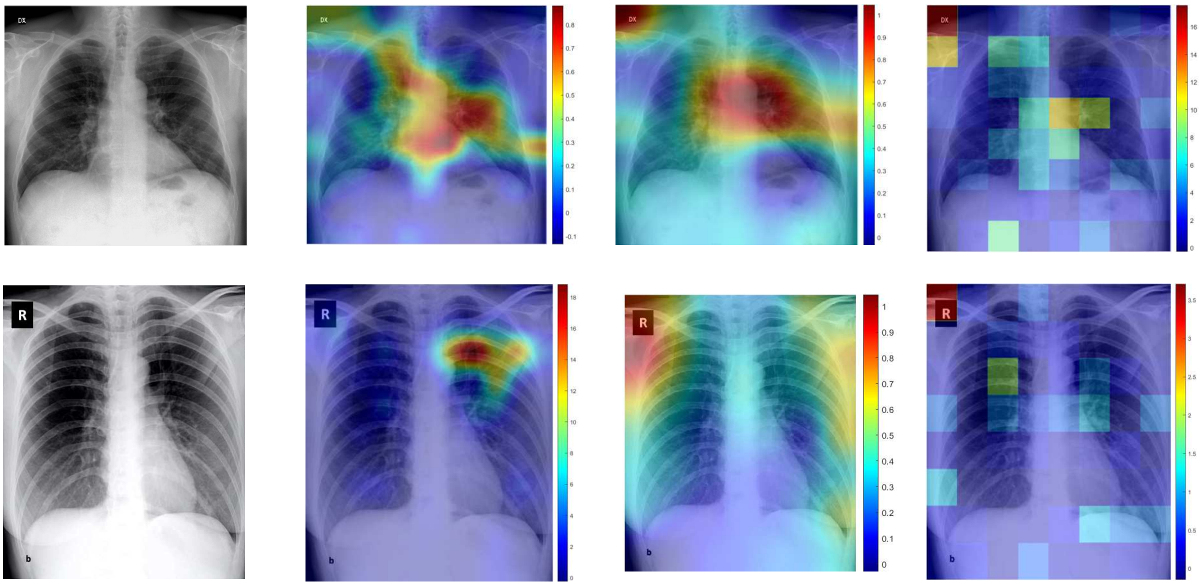


Fig. 5. Visualization of features: The first column shows the samples from the test set. The second, third and fourth column shows the heat maps from Occlusions sensitivity, GradCam, and LIME respectively.

models cannot fully substitute the prevailing testing methods until they achieve full explainability and transparency. There is an urgent need for advancement in the area of explainable AI, causability i.e. the quality of the explanations and also explanation interfaces. These models should provide a better understanding of the problem rather than just recognizing patterns. For now, AI models can be used to assist radiologists but not replace them.

This may not be the last pandemic. There are chances of future pandemics [24]. If patients feel normal while infectious, they may roam around normally and infect many before they are diagnosed. Coronavirus, though highly contagious, has a relatively low mortality rate. But a future outbreak may be more lethal, like Ebola. In such worst-case scenarios, rapid testing and identifying hotspots are the key steps, until a vaccine is developed. AI can be a solution for quick and large-scale screening and triaging patients for further procedures based on their risk assessment and severity.

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